ELECTRIC POWER STEERING (EPS) CONTROL MODULE

id061300245500

Purpose/ Function

- The EPS control module (CM) calculates the optimum assist current based on the steering torque signal from the torque sensor installed to the steering column and the vehicle speed signal sent via CAN transmission from the PCM.
- The following steering angle signals are output via CAN communication.
  - — Steering angle (relative angle) signal
    - Set the steering position to 0 degrees when the ignition is switched ON (engine off or on) to output the steering angle when steering from that position.
  - — Steering angle (estimated absolute angle) signal
    - The EPS CM outputs the calculated steering angle value to the CAN as a steering angle (estimated absolute angle) signal for other systems requiring steering angle information.

Note

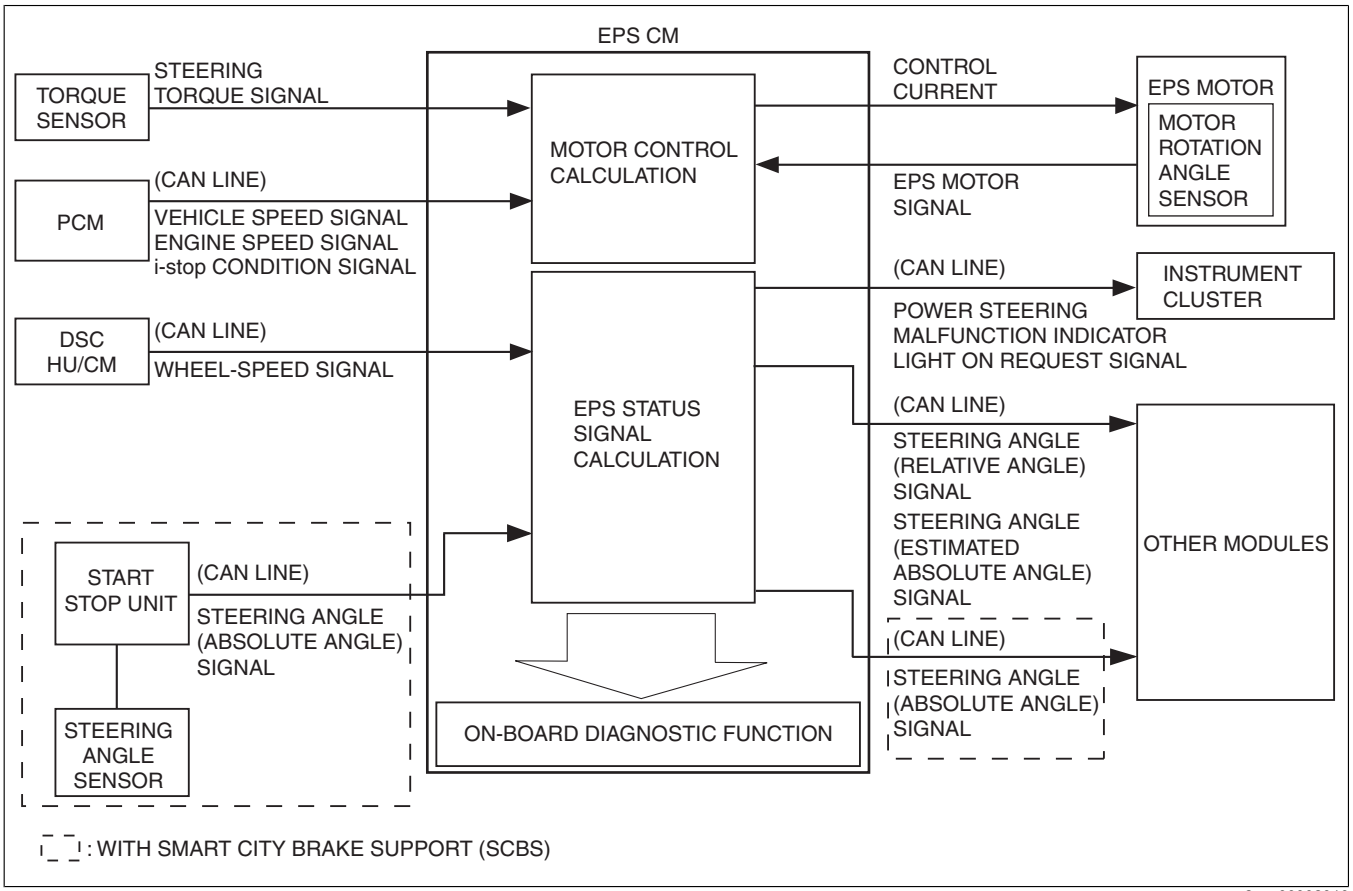
- The steering angle sensor signal is not used for the steering angle (estimated absolute angle) signal calculation.
- Before the steering wheel angle neutral position auto learning is completed, the EPS CM receives a steering angle (absolute angle) signal from the steering angle sensor via the start stop unit and outputs the signal to the other modules through the CAN communication (with Smart City Brake Support (SCBS)).
- After the steering angle neutral position auto learning is completed, the EPS CM outputs an EPS status signal, which is calculated by EPS CM itself, through the CAN communication.

- The EPS CM controls the following functions:

Function Table

Control item	Function
EPS motor current control	• Normal control — Calculates the proper assist current based on the steering torque, and the vehicle and engine speeds, and outputs a target current to the EPS motor. • System overheating protection control — Monitors the temperature of the motherboard in the EPS CM and controls the motor current gradually to prevent overheating of the system. • EPS motor backup control — Even if a phase 1 open circuit malfunction in the EPS motor occurs, output current to the EPS motor is controlled so that steering assist is maintained. • Motor rotation angle sensor backup control — Even if an open circuit malfunction occurs in the motor rotation angle sensor, the motor rotation angle sensor backup control performs control to maintain steering assist to the extent possible. — Monitors the temperature of the motherboard in the EPS CM and controls the motor current in steps to prevent overheating of the system. • EPS i-stop control — Control current to the EPS motor is increased or decreased according to the operation status of the i-stop system.
On-board diagnostic function	• The main part of the system control includes the self-diagnosis function. In case a malfunction occurs, the power steering malfunction indicator light illuminates or flashes to alert the driver, and a DTC is stored in the EPS CM at the same time. • As a result of the on-board diagnosis, when a malfunction is determined, system control is suspended or limited to assure safety while driving.
CAN communication function	• Outputs the EPS status signal and the power steering malfunction indicator light illuminates on request via CAN lines. • Outputs the steering angle (absolute angle) signal, steering angle (relative angle) signal and steering angle (estimated absolute angle) signal
Automatic configuration function	• When the ignition is switched to ON or the engine is started after the EPS CM have been replaced, the EPS CM reads data from the instrument cluster via CAN communication to perform automatic configuration.
Steering wheel angle neutral position automatic learning function	• When the ignition is switched from off to ON (engine on) and the vehicle is driven normally, the steering wheel angle neutral position is learned automatically based on the wheel speed signal, lateral-G signal, and yaw rate signal.

Block Diagram



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Construction

- The EPS CM is installed to the steering column and integrated with the EPS motor.
- For vehicles equipped with a system requiring steering angle information directly after the vehicle starts to be driven, a steering angle sensor is equipped.

Operation

- The EPS CM calculates the required assist torque based on the steering torque signal from the torque sensor. When the vehicle starts to be driven, the required assist torque is calculated based on the vehicle speed signal sent from the PCM via CAN in addition to the steering torque signal.
- When the required assist torque is determined, the EPS CM outputs the control current to the EPS motor and the EPS motor generates the assist torque.
- The EPS system control uses the steering angle (estimated absolute angle) signal.

  Ignition is switched ON (engine off or on)

  — For vehicles with a steering angle sensor, the EPS CM receives the steering angle sensor signal via the start stop unit. The EPS CM outputs the steering angle value as a steering angle (absolute angle) signal.

  — The following systems perform control using a steering angle (absolute angle) signal until the EPS CM completes steering wheel angle neutral position determination after the vehicle starts to be driven.
    - Smart City Brake Support (SCBS)

  — The EPS CM regards the steering position when the ignition is switched ON (engine off or on) as 0 degrees, and outputs the steering angle value steered from that position as a steering angle (absolute angle) signal.

  Vehicle driving

  — When the vehicle starts to be driven, the EPS CM performs steering wheel angle neutral position determination based on the following signals:
    - Wheel speed signal: Determines straight-driving if there is no rotation speed difference between the left/right wheels.

  — If the EPS CM completes steering wheel angle neutral position determination, the steering angle neutral position is set. The EPS CM regards the steering wheel angle neutral position as 0 degrees, calculates the estimated absolute steering angle, and outputs the steering angle (estimated absolute angle) signal to CAN.

  Ignition is switched off

  — The steering angle neutral position which the EPS CM determined automatically is cleared when **5 min or more** have elapsed after disconnecting the battery.

EPS Motor Current Control

Normal control

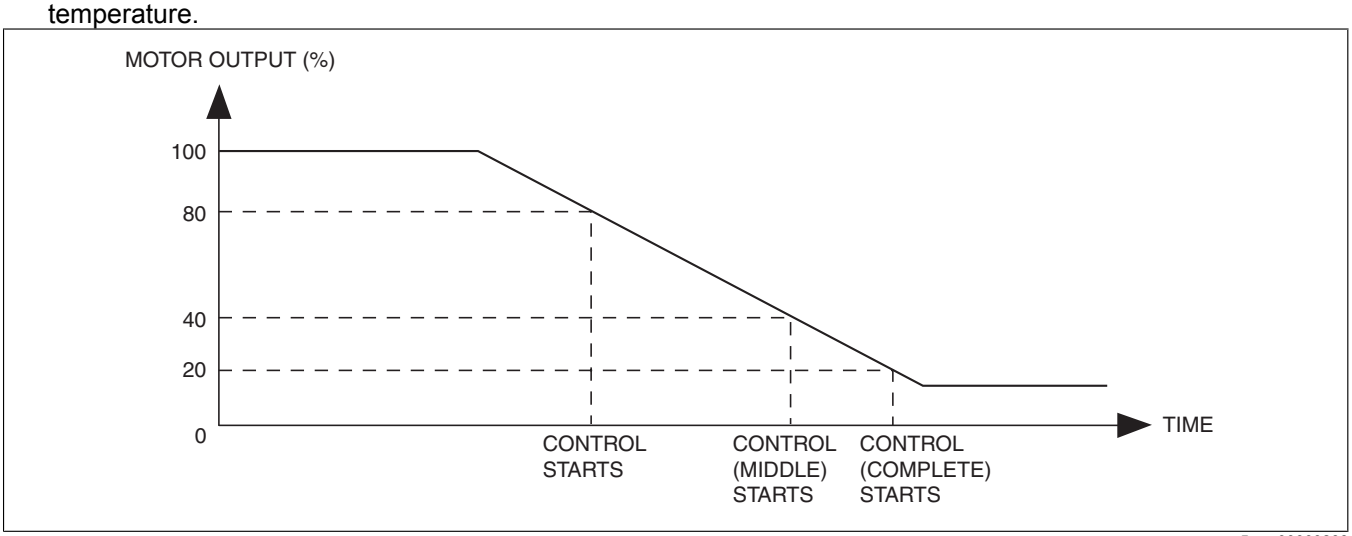
- The optimum assist current is calculated based on the steering force signal from the torque sensor and the vehicle and the engine speed signal from the PCM, and then the control current is output to the EPS motor.

System overheating protection control

- The system overheating protection control lowers the current output to the EPS motor if the steering mechanism is turned from lock to lock continuously or the steering rack turning limit is reached repeatedly.

  Note
  - During system overheating protection control, the assist force will feel weaker due to a decrease in the control current. This is normal for the EPS CM to prevent the EPS motor from burning or seizure, and it does not indicate a malfunction.
  - While the assist force weakens during this period of time, steering operation can be performed.

- The system overheating protection control has middle and complete steps, and it can verify current conditions and the history using the PID/data monitor function and snap shot data.
  - — System overheating protection control (middle): Motor output is 40 % or less (reference)
  - — System overheating protection control (complete): Motor output is 20 % or less (reference)
- The current output returns to normal if the temperature in the system decreases to the normal operation temperature.



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EPS motor backup control

- Even if an open circuit malfunction occurs in 1 out of the 3 phases in the EPS motor line, the EPS motor backup control performs control to maintain steering assist.
- If the EPS CM detects a phase 1 open circuit malfunction in the EPS motor, the EPS CM controls output current to the EPS motor using a calculation method different than during normal EPS CM control. As a result, even with a phase 1 open circuit malfunction, the steering assist can be maintained.

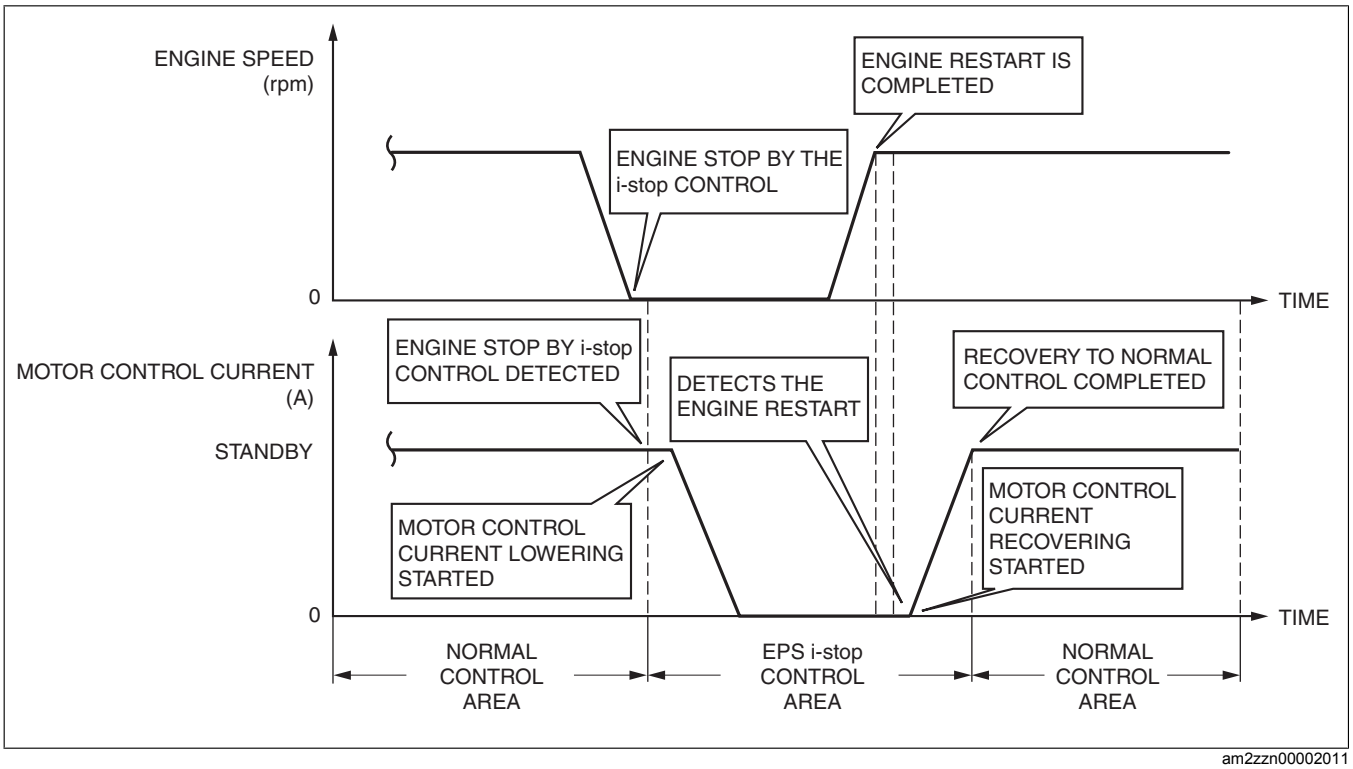
  Note
  - If the steering wheel is operated during EPS motor backup control, the steering wheel operation may feel heavier (resistance).

Motor rotation angle sensor backup control

- Even if an open circuit malfunction occurs in the motor rotation angle sensor, the motor rotation angle sensor backup control performs control to maintain steering assist.

EPS i-stop control

- If it is detected that the engine is stopped by the i-stop control based on the i-stop condition signal from the PCM, the EPS motor control current is stopped.
- If it is detected that the engine is restarted by the i-stop control, the EPS motor control current is increased and the system returns to normal control.
- If the steering wheel is operated during EPS i-stop control, assist is performed (normal control) according to that steering.
- If a malfunction is detected in the i-stop condition signal from the PCM, EPS i-stop control is not operated, and normal control is performed.
- Refer to [i-stop CONTROL] for a detailed explanation of the i-stop. (See i-stop CONTROL [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See i-stop CONTROL [SKYACTIV-D 1.5].)



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Fail-safe

DTC No.	Fail-safe function
U2300:54	• Control enabled • Power steering malfunction indicator light: Not illuminated
U2300:55	• Control enabled • Power steering malfunction indicator light: Illuminated
U2300:56	• Control enabled • Power steering malfunction indicator light: Not illuminated
U3000:05	• Control disabled • Power steering malfunction indicator light: Illuminated
U3000:41	
U3000:43	• Control enabled • Power steering malfunction indicator light: Not illuminated
U3000:46	• Control disabled • Power steering malfunction indicator light: Illuminated
U3000:49	
U3000:61	
U3000:96	
U3003:16	• Control is maintained by gradually decreasing the motor control current — However, control is inhibited if the power supply voltage is the specified value of less • Power steering malfunction indicator light: Illuminated
U3003:17	• Control disabled • Power steering malfunction indicator light: Illuminated
U3004:49	